U.G.C. NET Exan
COMPUTE
(Solved with Expla

1. **A hierarchical memory system that uses cache memory has cache access time of 50 nano seconds, main memory access time of 300 nano seconds, 75% of memory requests are for read, hit ratio of 0.8 for read access and the write-through scheme is used. What will be the average access time of the system both for read and write requests?**
   (a) 157.5 n.sec (b) 110 n.sec.
   (c) 75 n.sec. (d) 82.5 n.sec.

Ans : (a) Because, $T_{read} = 0.8 \times 50 + 0.2 \times (300 + 50) = 110$ NS
$T_{write} = 1 \times \max(300, 50) = 1 \times 300 = 300$ ns
Hence average access time for both read and write is = $0.75 \times 110 + 0.25 \times 300 = 157.5$ ns

2. **For switching from a CPU user mode to the supervisor mode following type of interrupt is most appropriate :**
   (a) Internal interrupts (b) External interrupts
   (c) Software interrupts (d) None of the above

Ans : (c) Because, internal interrupts occur due to some internal error (like division of zero)
External interrupts occur due H/w problem
Software interrupts are the most appropriate method for switching user mode to kernel mode.

3. **In a dot matrix printer the time to print a character is 6 m.sec., time to space in between characters is 2 m.sec., and the number of characters in a line are 200. The printing speed of the dot matrix printer in characters per second and the time to print a character line are given by which of the following options?**
   (a) 125 chars/seconds and 0.8 seconds
   (b) 250 chars/seconds and 0.8 seconds
   (c) 166 chars/second and 0.8 seconds
   (d) 250 chars/second and 0.4 seconds

Ans : (*) Because, total characters = 200
total time to print 200 chars. = $200 \times 6 = 1200$ ms
total time to space in = $200 \times 2 = 400$ ms
200 characters total print time = 1600 ms = 1.6 sec
The printing speed of the dot matrix printer is character per second = 200/1.6 = 125 chars/sec.

4. **Match the following 8085 instructions with the flags :**

List – I		List – II	
A.	**XCHG**	**i.**	**only carry flag is affected**
B.	**SUB**	**ii.**	**no flags are affected**
C.	**STC**	**iii.**	**all flags other than carry flag are affected**
D.	**DCR**	**iv.**	**all flags are affected**

Code :

	A	B	C	D
(a)	iv	i	iii	ii
(b)	iii	ii	i	iv
(c)	ii	iii	i	iv
(d)	ii	iv	i	iii

Ans : **(d)** Because, (a) XCHG : Exchange H & L with D & E no flag is affected in this instruction
(b) SUB : Subtract all flags (CY, S, Z, AC, P) are affected by subtract operation
(c) STC : Set Carry flag to 1 so only carry flag is affected
(d) DCR : Decrement all flags other than carry flag are affected.
Hence the answer is (d)

5. **How many times will the following loop be executed?**
LOP : DCX B
MOV A, B
ORA C
JNZ LOP
(a) 05 (b) 07
(c) 09 (d) 00

Ans : **(b)** Because, value 7 is placed in register pair B every time value is by one by DCX B
Contents of B are moved to accumulator and logical OR is performed with C but that don't affect the repetition of loop which is controlled by JNZ(JUMP is not zero) LOP instruction. So loop executes for B values 7, 6, 5, 4, 3, 2, 1 total 7 times

6. **Specify the contents of the accumulator and the status of the S, Z and CY flags when 8085 microprocessor performs addition of 87 H and 79 H.**
(a) 11, 1, 1, 1 (b) 10, 0, 1, 0
(c) 01, 1, 0, 0 (d) 00, 0, 1, 1

Ans : (d) Because, 87 H – 10000111

79 H – 01111001

100000000

S–sign flag– Since MSB of result is 0, hence sign bit should be 0.

Z-zero flag– Since result is zero, hence zero bit should be 1.

Cy–Carry flag– Since these is carry from MSB, hence Cy = 1

7. Location transparency allows :

I. Users to treat the data as if it is done at one location

II. Programmers to treat data as if it is at one location

III. Managers to treat the data as if it is at one location

Which of the following is correct?

(a) I, II and III (b) I and II only

(c) II and III only (d) II only

Ans : (a) Because, In distributed computer networks, location transparency is the use of names to identify network resources, rather than their actual location.

8. Which of the following is correct?

I. Two phase locking is an optimistic protocol

II. Two phase locking is pessimistic protocol

III. Time stamping is an optimistic protocol

IV. Time stamping is pessimistic protocol

(a) I and III (b) II and IV

(c) I and IV (d) II and III

Ans : (d) Two phase locking is pessimistic protocol (assumes bad thing going to happen) since it applies locks (for safety)

Time stamping is an optimistic protocol (assumes good thing going to happen) since it uses no locking. If bad things happen, then they are handled later.

9. ....rules used to limit the volume of log information that has to be handled and processed in the event of system failure involving the loss of volatile information:

(a) Write-ahead log (b) Check-pointing

(c) Log buffer (d) Thomas

Ans : (b) Check pointing is a technique to add fault tolerance into computer system. It basically consist of saving a snapshot of the application state, so that it can restart from that point in case of failures

10. Let R = ABCDE is a relational scheme with functional dependency set F = {A → B, B → C, AC → D}. The attribute closures of A and E are :

(a) ABCD, ϕ (b) ABCD, E

(c) Φ, ϕ (d) ABC, E

Ans : (b) {A}+ = {A, B, C, D}
{F} = {E}
Hence the correct option is (b)

11. Consider the following statements :

I. Re-construction operation used in mixed fragmentation commutative rule.

II. Re-construction operation used in vertical fragmentation satisfies commutative rule

Which of the following is correct?

(a) I
(b) II
(c) Both are correct
(d) None of the statements are correct

Ans : (d) All reconstructions are need two operation
- Join
- Union

Because the operation above are commutative, the reconstructions are commutative too.

12. Which of the following is false ?

(a) Every binary relation is never be in BCNF
(b) Every BCNF relation is in 3NF
(c) 1 NF, 2 NF, 3 NF and BCNF are based on functional dependencies
(d) Multivalued dependency (MVD) is a special case of join dependency (JD)

Ans : (a) Every binary relation (a relation with only 2 attributes) is always in BCNF. Because if both attributes form the primary key, the relation is in BCNF. If one of the attribute is a primary key, the other must be determined by it and thus in BCNF.

13. Which of the following categories of language do not refer to animation language?

(a) Graphical languages
(b) General-purpose languages
(c) Linear-list notations
(d) None of the above

Ans : (d) All 3 mentioned are refer to animation language. Linear list notations are B rotate "Palm" etc. General purpose language eg. Grasp my-cube etc.
General language describe animation in a usual way instead of using a scripts.
So the correct choice is (d)

14. Match the following :

	List –I		List –II
A.	**Tablet, Joystick**	**i.**	**Continuous devices**
B.	**Light pen, Touch screen**	**ii.**	**Direct devices**
C.	**Locator, Keyboard**	**iii.**	**Logical devices**
D.	**Data Globe, Sonic pen**	**iv.**	**3D interaction devices**

Codes:

	A	B	C	D
(a)	ii	i	iv	iii
(b)	i	iv	iii	ii
(c)	i	ii	iii	iv
(d)	iv	iii	ii	i

Ans : (c) Keyboard is a logical device & so is locator data globe, sonic pen are 3D interaction devices

Table joystick are continuous devices as we can not lift our hand during their operation

Light Pen & touch screen are direct devices as we directly touch them & get our work done.

15. A technique used to approximate halftones without reducing spatial resolution is known as............

(a) Half toning (b) Dithering

(c) Error diffusion (d) None of the above

Ans : (b) It is a technique can be used to approximate halftones without reducing spatial resolution. In this approach the dither matrix is treated very much like a floor file that can be repeatedly positioned one copy next to another to cover entire floor.

16. Consider a triangle represented by A(0,0), B(1,1), C(5,2). The triangle is rotated by 45 degrees about a point P(–1, –1). The co-coordinates of the new triangle obtained after rotation shall be–

(a) $A'\left(-1,\sqrt{2}-1\right), B'\left(-1,2\sqrt{2}-1\right), C''\left(\frac{3}{2}\sqrt{2}-1,\frac{9}{2}\sqrt{2}-1\right)$

(b) $A'\left(\sqrt{2}-1,-1\right), B'\left(2\sqrt{2}-1,-1\right), C''\left(\frac{3}{2}\sqrt{2}-1,\frac{9}{2}\sqrt{2}-1\right)$

(c) $A'\left(-1,\sqrt{2}-1\right), B'\left(2\sqrt{2}-1,-1\right), C''\left(\frac{3}{2}\sqrt{2}-1,\frac{9}{2}\sqrt{2}-1\right)$

(d) $A'\left(-1,\sqrt{2}-1\right), B'\left(2\sqrt{2}-1,-1\right), C''\left(\frac{3}{2}\sqrt{2}-1,\frac{9}{2}\sqrt{2}-1\right)$

Ans : (a)

$$R_{45}, p = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 1 & \left(\sqrt{2}-1\right) & 1 \end{pmatrix}$$

Now,

$$\begin{bmatrix} A' \\ B' \\ C' \end{bmatrix} = \begin{bmatrix} A \\ B \\ C \end{bmatrix} R_{45}p = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 5 & 2 & 1 \end{pmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -1 & \left(\sqrt{2}-1\right) & 1 \end{bmatrix}$$

$$= \begin{pmatrix} -1 & \left(\sqrt{2}-1\right) & 1 \\ -1 & \left(2\sqrt{2}-1\right) & 1 \\ \left(\frac{3}{2}\sqrt{2}-1\right) & \left(\frac{9}{2}\sqrt{2}-1\right) & 1 \end{pmatrix}$$

So $A' = \left(-1, \sqrt{2}-1\right), B' = \left(-1, 2\sqrt{2}-1\right)$ & $C' = \left(\frac{3}{2}\sqrt{2}-1, \frac{9}{2}\sqrt{2}-1\right)$

Hence the correct choice is (a)

17. In Cyrus-Beck algorithm for line clipping the value of t parameter is computed by the relation :
(Here P_1 and P_2 are the two end points of the line, f is a point on the boundary, n_1 is inner normal)

(a) $\frac{\left(P_1 - f_i\right).n_i}{\left(P_2 - P_1\right).n_i}$ (b) $\frac{\left(f_i - P_1\right).n_i}{\left(P_2 - P_1\right).n_i}$

(c) $\frac{\left(P_2 - f_i\right).n_i}{\left(P_1 - P_2\right).n_i}$ (d) $\frac{\left(f_i - P_2\right).n_i}{\left(P_1 - P_2\right).n_i}$

Ans : (b) $\frac{\left(f_i - P_1\right).n_i}{\left(P_2 - P_1\right).n_i}$

18. Match the following :

A.	**Cavalier Projection**	**i.**	**The direction of projection is chosen so that there is no foreshortening of lines perpendicular to the xy plane**
B.	**Cabinet Projection**	**ii.**	**The direction of projection is chosen so that lines perpendicular to the xy planes are foreshortened by half their lengths**
C.	**Isometric Projection**	**iii.**	**The direction of projection makes equal angles with all of the principal axis**

D.	**Orthographic Projection**	**iv.**	**Projections are characterized by the fact that the direction of projection is perpendicular to the view plane**

Code :

	A	B	C	D
(a)	i	iii	iv	ii
(b)	ii	iii	i	iv
(c)	iv	ii	iii	i
(d)	i	ii	iii	iv

Ans : (d) **Cavalier projection :** It sometimes called cavalier perspective or high view point. A point of the object is represented by three coordinates x,y & z. On the drawing, it is represented by only two coordinates "x" & "y" on the flat drawing, two axes, x & z on the figure, are perpendicular to the view of plane.

(b) Cabinet projection : Like, cavalier, one face of the project object is parallel to the viewing plane, In this the third axis keeps its length, with cabinet projection of the length of the receding lines is cut in half

(c) Isometric projection : In this reflecting that the scale along each axis of the projection is the same.

(d) Orthographic projection : It is a means of representing a three dimensional object in two dimension it is a form of parallel projection.

19. Consider the following statements S1, S2 and S3 :

S1 : In call–by–value, anything that is passed into a function call is unchanged in the caller's scope when the function returns.

S2 : In call-by-reference, a function receives implicit reference to a variable used as argument.

S3 : In call-by-reference, caller is unable to see the modified variable used as argument

(a) S3 and S2 are true
(b) S3 and S1 are true
(c) S2 and S1 are true
(d) S1, S2, S3 are true

Ans : (c) **Call-by-value :** While passing parameters using call by value, copy of original parameter is created & passed to the called function & any update made inside method will not effect the original value variable in calling function.

Call-by-reference : This method copes the address of an argument into the formal parameter.

So answer is c, because in call-by reference caller always sees the modified value of the argument.

20. How many tokens will be generated by the scanner for the following statement?
x = x * (a + b) –5;

(a) 12 (b) 11
(c) 10 (d) 07

Ans : **(**a) x = x × (a + b) – 5
Every character is taken here

Token no	character
1	x
2	=
3	x
4	*
5	(
6	a
7	+
8	b
9	)
10	–
11	5
12	;

21. Which of the following statements is not true?

(a) MPI_Isend and MPI_Irecv are non-blocking message passing routines of MPI
(b) MPI_Issend and MPI_Ibsend are non-blocking message passing routines of MPI
(c) MPI_Send and MPI_Recv are non-blocking message passing routines of MPI
(d) MPI_Ssend and MPI_Bsend are blocking message passing routines of MPI

Ans : (c) Message-passing interface (MPI) standard. MPI_Isend & MPI_Irecv are non-blocking MPI_Isend buffered non blocking so statement A & B are correct & MPI_Send & MPI_Bsend are blocking message, S, D is also correct statement
So the correct answer is (c)

22. The pushdown automation M = ($\{q_0, q_1, q_2\}$, {a, b}, {0, 1}, δ, q_0, 0, $\{q_0\}$) with
$\delta(q_0, a, 0) = \{(q_1, 10)\}$
$\delta(q_1, a, 1) = \{q_1, 11)\}$
$\delta(q_1, b, 1) = \{q_2, \lambda)\}$
$\delta(q_2, b, 1) = \{q_2, \lambda)\}$
$\delta(q_2, \lambda, 0) = \{q_0, \lambda)\}$
Accepts the language

(a) $L = \{a^n b^m | n, m \geq 0\}$
(b) $L = \{a^n b^n | n \geq 0\}$
(c) $L = \{a^n b^m | n, m > 0\}$
(d) $L = \{a^n b^n | n > 0\}$

Ans : (b) This PDA clearly accepts a^n b^n as its language but as we see that there is a production from Q_2 to q_0 makes the input string half on Q_0, So out final state will be Q_0 So, it will accept empty strings.

23. Given two languages :

$L1 = \{(ab)^n a^k \mid n > k, k \geq 0\}$

$L2 = \{a^n b^m \mid n \neq m\}$

Using pumping lemma for regular language, it can be shown that

(a) L_1 is regular and L_2 is not regular

(b) L_1 is not regular and L_2 is regular

(c) L_1 is regular and L_2 is regular

(d) L_1 is not regular and L_2 is not regular

Ans : (d) Whenever pumping lemma is there it can only be used to prove that language is not regular or not context free

So the answer is (d)

24. Regular expression for the complement of language $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$ is :

(a) $(a + b) \times ba (a + B) \times$

(b) $A \times bbbbb \times$

(c) $(\lambda + a + aa + aaa)b \times + (a + b) \times ba(a + b)\times$

(d) None of the above

Ans : (d) $n \geq 4$, then {aaaa, aaaaa, aaaaaa, aaaaaaa.....}

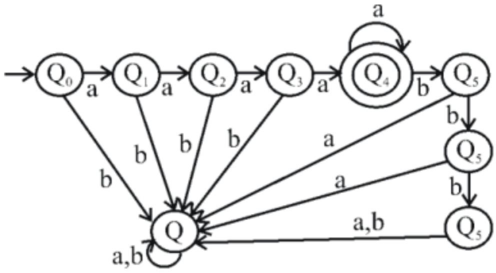
= aaaa {E, a, aa, aaa, ...} = aaaaa*

$m \leq 3$, then {E, b, bb, bbb}

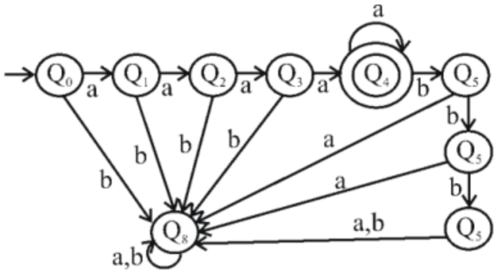
$L = \{a^n b^m \mid n \geq 4, m \leq 3\}$

Regular expression will be aaaaa* (E + b + bb + bbb)

DFA for above regular expression will be



DFA for complement of L, i.e. L will be



Will regular expression

(E + a + aa + aaa) (E + b) (a + b)*) + aaaaa* (b + bb) a (a + b)* + aaaaa* bbb (a + b) (a + b)*

So the correct choice is (d)

25. For n devices in a network, ... number of duplex-mode links are required for a mesh topology.

(a) $n(n + 1)$ (b) $n(n - 1)$

(c) $n(n + 1)/2$ (d) $n(n - 1)/2$

Ans : (d) In mesh topology, every computer in the network has a connection to each of the other computers in that network. The number of connections in this network can be calculated using the following formula n(n–1)/2 (where n is the number of computer in the network)

26. How many characters per second (7 bits + 1 parity) can be transmitted over a 3200 bps line if the transfer is asynchronous? (Assuming 1 start bit and 1 stop bit)

(a) 300 (b) 320
(c) 360 (d) 400

Ans : (b) In asynchronous transmission, for each character, we need to send data, parity, start & stop bits.
size of one character = 7 + 1 + 1 + 1 = 10 bits
Line transmits 3200 bits in one second.
no. of characters transmitted in one second = 3200/10 = 320

27. Which of the following is not a field in TCP header?

(a) Sequence number (b) Fragment offset
(c) Checksum (d) Window size

Ans : (b)

<table>
<tr><td>Bit 0</td><td></td><td>Bit 15</td><td>Bit 16   Bit 31</td></tr>
<tr><td colspan="3">Source part (16)</td><td>Destination part (16)</td></tr>
<tr><td colspan="4">Sequence number (32)</td></tr>
<tr><td colspan="4">Acknowledgment number (32)</td></tr>
<tr><td>Header length (4)</td><td>Reserved (6)</td><td>Code bits (6)</td><td>Window (16)</td></tr>
<tr><td colspan="3">Checksum (16)</td><td>Urgent (16)</td></tr>
<tr><td colspan="4">Option (0-32 if any)</td></tr>
<tr><td colspan="4">Data (varies)</td></tr>
</table>

TCP Header

28. What is the propagation time if the distance between the two point is 48,000? Assume the propagation speed to be 2.4×10^8 meter/second in cable :

(a) 0.5 ms (b) 20 ms
(c) 50 ms (d) 200 ms

Ans : (d) Distance between points = 48000 km = 48×10^6 meter
Propagation speed = 2.4×10^8 meter/sec

$$\text{Propagation time} = \frac{\text{Distance between points}}{\text{Propagation speed}}$$

$$= \frac{48\times10^6}{2.4\times10^8} = 0.2\text{ sec} = 200\text{ ms}$$

29. .... is a bit-oriented protocol for communication over point-to-point and multipoint links.

(a) Stop-and-wait (b) HDLC
(c) Sliding window (d) Go-back-N

Ans : (b) HDLC (High level data link control) is a classical bit oriented code transparent synchronous data link layer protocol by ISO protocol.
HDLC can be used for point to multipoint connections. But is now used almost exclusively to connect the one device to another.

30. Which one of the following is true for asymmetric-key cryptography?

(a) Private key is kept by the receiver and public key is announced to the public
(b) Public key is kept by the receiver and private key is announced to the public
(c) Both private key and public key are kept by the receiver
(d) Both private key and public key are announced to the public

Ans : (a) In asymmetric key cryptography uses two keys private key & public key. Private key is kept by the receiver & public key is announced to the public
So answer is (a)

31. Any decision tree that sorts n elements has height

(a) $\Omega(n)$ (b) $\Omega(\lg n)$
(c) $\Omega(n \lg n)$ (d) $\Omega(n^2)$

Ans : (c) For any comparison based sorting algorithm the decision will have n! leaf nodes, so height will be $\log(n!) = \Omega(n \log n)$

32. Match the following :

List –I		**List –II**	
A.	**Bucket sort**	**i.**	$O(n^3 \lg n)$
B.	**Matrix chain multiplication**	**ii.**	$O(n^3)$
C.	**Huffman codes**	**iii.**	$O(n \lg n)$
D.	**All pairs shortest paths**	**iv.**	$O(n)$

Codes :

	A	**B**	**C**	**D**
(a)	iv	ii	i	iii
(b)	ii	iv	i	iii
(c)	iv	ii	iii	i
(d)	iii	ii	iv	i

Ans : (c)
(a) Bucket sort : Bucket sort is mainly useful when input is uniformly over a range. Time complexity is o(n) time.
(b) Matrix chain multiplication : In this no matter how we parenthesize the product, the result will be same & the time complexity is $o(n^3)$ time
(c) Huffman code : It is a lossless data compression algorithm. The time complexity is o(nlgn) times
(d) All pairs shortest paths : The Floyd warshall algorithm is solving the all pairs shortest path problem.

The problem is to find shortest distance between every pair of vertices and the time complexity is $o(n^3 lgn)$.

33. We can show that the clique problem is NP-hard by proving that :

(a) CLIQUE ≤ P 3–CNF_SAT
(b) CLIQUE ≤ P VERTEX_COVER
(c) CLIQUE ≤ P SUBSET_SUM
(d) None of the above

Ans : (d) **Clique :** A clique in graph G is a complete graph; that is, it is a subset S of the vertices such that every two vertices is S are connected by an edge in G the decision problem is a NP complete.

3 SAT : SAT asks whether the variable of a given Boolean formula can be consistently replaced by the values TRUE or FALSE is such way that the formula evaluates to TRUE SAT is one of the first problems that was proven to be NP-complete & vertex problem & subset sum are NP-complete problem. So option (a, b, c) match with rule (3, 4)

So the correct answer is (d)

34. Dijkstra algorithm, which solves the single-source shortest-paths problem, is a ...., and the Floyd-Warshall algorithm, which finds shortest paths between all pairs of vertices, is a...........

(a) Greedy algorithm, Divide-conquer algorithm
(b) Divide-conquer algorithm, Greedy algorithm
(c) Greedy algorithm, Dynamic programming algorithm
(d) Dynamic programming algorithm, Greedy algorithm

Ans : (c) Dijkstra algorithm, which solves the single source shortest path problem, is a greedy algorithm & the Floyd warshall algorithm, which finds shortest paths between all pair of vertices, is a dynamic programming algorithm.

35. Consider the problem of a chain <A1, A2, A3 > of three matrices. Suppose that the dimensions of the matrices are 10 × 100, 100 × 5 and 5 × 50 respectively. There are two different ways of parenthesization : (i) ((A1 A2)A3) and (ii) (A1 (A2A3)). Computing the product according to the first parenthesization is .... times faster in comparison to the second parenthesization.

(a) 5 (b) 10
(c) 20 (d) 100

Ans : (b) Assume one multiplication take 1 m sec.

I. will take 7500 multiplication = 7500 × 1 msec = 7500 msec

II. will take 7500 multiplication = 75000 × 1 msec = 75000 msec faster = 75000/7500 = 10

So answer is (b)

36. Suppose that we have numbers between 1 and 1000 in a binary search tree and we want to search for the number 365. Which of the following sequences could not be the sequence of nodes examined?

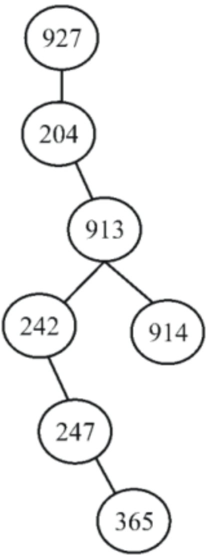
(a) 4, 254, 403, 400, 332, 346, 399, 365

(b) 926, 222, 913, 246, 900, 260, 364, 365

(c) 927, 204, 913, 242, 914, 247, 365

(d) 4, 401, 389, 221, 268, 384, 283, 280, 365

Ans : (c)



Option (c) is wrong because 914 can be traveled after 913

37. Which methods are utilized to control the access to an object in multi-threaded programming?

(a) A synchronized methods

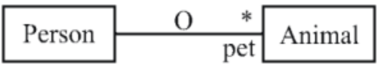
(b) Synchronized methods

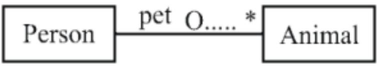
(c) Serialized methods

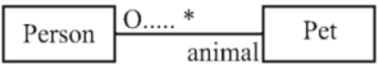
(d) None of the above

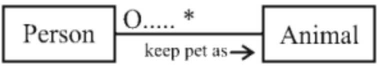
Ans : (b) Synchronation is the capability to control the access of multiple threads to shared resources, It is possible for one java thread to modify a shared variable while another thread is in the process of using or updating same shared variable.

38. How to express that some person keeps animals as pets?

(a) 

(b) 

(c) 

(d) 

Ans : (a) A person may have none or many animals as pets (one to many relationships) which is shown by (a), Hence the correct answer is (a)

39. Converting a primitive type data into its corresponding wrapper class object instance in called

(a) Boxing (b) Wrapping

(c) Instantiation (d) Auto boxing

Ans : (d) Autoboxing is the automatic conversion that the java compiler makes between the primitive types & their corresponding object wrapper class.

40. The behavior of the document elements in XML can be defined by :

(a) Using document object

(b) Registering appropriate event handlers

(c) Using element object

(d) All of the above

Ans : (b) Java script program can define the document elements by registering appropriate event handless.

41. What is true about UML stereotypes?

(a) Stereotype is used for extending the UML language

(b) Stereotyped class must be abstract

(c) The stereotype indicates that the UML element cannot be changed

(d) UML Profiles can be stereotyped for backward compatibility

Ans : (a)

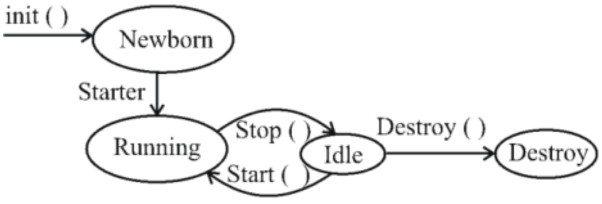
- Stereotype, tags & constraints are used for extending the UML
- They allow designers to extend the vocabulary of UML in order to create new model elements, derived from existing ones.
- Stereotype is rendered as a name enclosed by << >>
- Stereotype cannot be extends by another stereotype

42. Which method is called first by an applet program?

(a) start ( ) (b) run ( )

(c) init ( ) (d) begin ( )

Ans : (c) applet life cycle



So the answer is (c)

43. Which one of the following is not a source code metric?

(a) Halstead metric

(b) Function point metric

(c) complexity metric

(d) Length metric

Ans : (b) Function point metric, this approach is independent of the language tools or methodologies used for implementation.

44. To compute function points (FP), the following relationship is used

FP = Count – total × (0.65 + 0.01 × Σ(F_i) where Fi (i = 1 to n) are value adjustment factors (VAF) based on n questions. The value of n is :

(a) 12 (b) 14

(c) 16 (d) 18

Ans : (b) Number of value adjustment factor are based on response of 14 (Fi, i = 1 to 14) question

Fp = count total x (0.65 + 0.01 × Σ (Fi)) where count total is the sum of all FP entries.

45. Assume that the software term defines a project risk with 80% probability of occurrence of risk in the following manner:

Only 70 percent of the software components scheduled for reuse will be integrated into the application and the remaining functionality will have to be custom developed. If 60 reusable components were planned with average component size as 100 LOC and software engineering cost for each LOC as $ 14, then the risk exposure would be :

(a) $ 25,200 (b) $ 20,160

(c) $ 17,640 (d) $ 15,120

Ans : (b) Number of reusable components planed = 60

- Part of reusable component actually included = 70%
- Number of components developed freshly = 30% of 60 = 18
- Development cost = 18 × 100 × 14%
- Probability of occurrence risk = 80%
- Risk exposure = 0.8 × 18 × 100 × 14 = 20160$

So the answer is (b)

46. Maximum possible value of reliability is :

(a) 100 (b) 100

(c) 1 (d) 0

Ans : (c) Maximum possible reliability is 1 or 100%

So answer is (c)

47. 'FAN IN' of component A is defined as

(a) Count of the number of components that can call, or pass control, to a component A

(b) Number of components related to component

(c) Number of components dependent on component

(d) None of the above

Ans : (a) FAN-in is the number of modules that call a given module

Fan-out is the number of modules that called by a given module.

48. Temporal cohesion means

(a) Coincidental cohesion
(b) Cohesion between temporary variables
(c) Cohesion between local variables
(d) Cohesion with respect to time

Ans : (d) Temporal cohesion is when pairs of module is grouped by when they are processed the parts are processed at a particular time in program execution.

49. Various storage devices used by an operating system can be arranged as follows in increasing order of accessing speed :

(a) Magnetic tapes → magnetic disks → optical disks → electronic disks → main memory → cache → registers
(b) Magnetic tapes → magnetic disks → electronic disks → optical disks → main memory → cache → registers
(c) Magnetic tapes → electronic disks → magnetic disks → optical disks → main memory → cache → registers
(d) Magnetic tapes → optical disks → magnetic disks → electronic disks → main memory → cache → registers

Ans : (d) The electronics are definitely faster because they are ram or flash. So the answer is (d)

50. How many disk blocks are required to keep list of free disk blocks in a 16 GB hard disk with 1 kb block size using linked list of free disk blocks? Assume that the disk block number is stored in 32 bits.

(a) 1024 blocks (b) 16794 blocks
(c) 20000 blocks (d) 1048576 blocks

Ans : (none of these**)**

256 disk block address → 1 disk block

16 M disk block address → X

$$X = \frac{16M}{256} = 65794 \qquad \{M \rightarrow 2^{20}\}$$

51. Consider an imaginary disk 40 cylinders. A request come to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders1, 36, 16, 34, 9 and 12 in that order. The number of arm motions using shortest seek first algorithm is :

(a) 111 (b) 112
(c) 60 (d) 61

Ans : (d) 11 – 12 – 9 – 16 – 34 – 36 sequences of access

So, 1 + 3 + 7 + 15 + 33 + 2 = 61

Hence the answer is (d)

52. **An operating system has 13 tape drives. There are three processes P1, P2 & P3. Maximum requirement of P1 is 11 tape drives, P2 is 5 tape drives and P3 is 8 tape drives. Currently, P1 is allocated 6 tape drives, P2 is allocated 3 tape drives and P3 is allocated 2 tape drives. Which of the following sequences represent a safe state?**

(a) P2 P1 P3 (b) P2 P3 P1
(c) P1 P2 P3 (d) P1 P3 P2

Ans : (a)

Allocated	Max	need
P1 6	11	5
P2 3	5	2

Total resources = 13 & free resources = 13 – 11 = 2

So clearly these 2 can only be given

allocated	max	need (20)
P3 1	8	6

to P2 which will be completed 8 release its all 5 resource then next P1 can take 5 resource & be completed & finally P3

So right answer is (a) P2 P1 P3

53. **Monitor is an Interprocess Communication (IPC) technique which can be described as :**

(a) It is higher level synchronization primitive and is a collection of procedures, variables, and data structures grouped together in a special package
(b) It is a non-negative integer which apart from initialization can be acted upon by wait and signal operations
(c) It uses two primitives, send and receive which are system calls rather than language constructs
(d) It consists of the IPC primitives implemented as system calls to block the process when they are not allowed to enter critical region to save CPU time

Ans : (a) Monitor is a thread-safe class, object or module that uses wrapped mutual exclusion in order to safely allows access to a method or variable by more than one thread monitor is that its methods are executed with mutual exclusion : at each point in time & monitor grouped their data's in a special package.

54. **In a distributed computing environment, distributed shared memory is used which is :**

(a) Logical combination of virtual memories on the nodes
(b) Logical combination of physical memories on the nodes
(c) Logical combination of the secondary memories on all the nodes
(d) All of the above

Ans : (b) DSM (Distributed shared memory) is a form of memory architecture where the (physical separate) memories can be addressed as one (logical shared) address space. DSM system is to make interprocess communication transparent to end users.

55. Equivalent logical expression for the Well Formed Formula (WFF) $\sim(\forall x)F[x]$ is :

(a) $\forall x\,(\sim F[x])$ (b) $\sim(\exists x)F[x]$
(c) $\exists x(\sim F[x])$ (d) $\forall x F[x]$

Ans : **(c)** $\sim((\forall x)\, fx) = (\exists X)\,(\sim F(x))$
So answer is (c)

56. An A* algorithm is a heuristic search technique which :

(a) is like a depth-first search where most promising child is selected for expansion
(b) generates all successor nodes and computes an estimate of distance (cost) form start node to a goal node through each of the successors. It then chooses the successor with shortest cost
(c) saves all path lengths (costs) from start node to all generated nodes and chooses shortest path for further expansion
(d) none of the above/र

Ans : **(b)** A* is an informed search algorithm, or a best-first search, meaning that it solves problems by searching among all possible paths to the solution for the one that incurs the smallest cost (least distance travelled, shortest time etc) and among these paths it first consider the ones that appear to lead most quickly to the solution.

57. The resolvent of the set of clauses $(A \vee B, \sim A \vee D, C \vee \sim B)$ is :

(a) $A \vee B$ (b) $C \vee D$
(c) $A \vee C$ (d) $A \vee D$

Ans : **(b)** $A\vee B, \sim A\vee D, C\vee\sim B$
A & ~A will cancel out & so will B & ~B
Hence answer is (b)

58. Match the following :

A.	**Script**	**i.**	**Directed graph with labelled nodes for graphical representation of knowledge**
B.	**Conceptual Dependencies**	**ii.**	**Knowledge about objects and events is stored in record-like structures consisting of slots and slot values**

C.	**Frames**	**iii.**	**Primitive concepts and rules to represent natural language statement**
D.	**Associative Network**	**iv.**	**Frame like structures used to represent stereotypical patterns for commonly occurring events in terms of actors, roles, props and scenes**

Codes :

	A	B	C	D
(a)	iv	ii	i	iii
(b)	iv	iii	ii	i
(c)	ii	iii	iv	i
(d)	i	iii	iv	i

Ans : (b)

(a) Script : Scripts are a development of the idea of representing action and events using semantic networks. Script is the idea is that whole set of action fall into stereotypical patterns.

(b) Conceptual Dependences : Conceptual dependency is a model of natural language understanding used in AI system means it is a primitive concepts & rules to represent natural language statements.

(c) Frames : consideration of the use of cases suggest how we can tighten up on the semantic net notation to give something which is more consistent.

(d) Associative network : Associate networks are cognitive models that incorporate long-known principles of association to represent key feature of memory means it is a directed graph with labelled nodes for graphical representation of knowledge.

59. Match the following components of an expert system :

A.	**I/O interface**	**i.**	**Accepts user's queries and responds to question through I/O interface**
B.	**Explanation module**	**ii.**	**Contains facts and rules about the domain**
C.	**Inference engine**	**iii.**	**Gives the user, the ability to follow**

			inferencing steps at any time during consultation
D.	**Knowledge base**	**iv.**	**Permits the user to communicate with the system in a natural way**

Codes :

	A	B	C	D
(a)	i	iii	iv	ii
(b)	iv	iii	i	ii
(c)	i	iii	ii	iv
(d)	iv	i	iii	ii

Ans : (b)
(a) I/O interface : Method is used to transfer information between internal storage & external I/O device is know as I/O interface means it permits the user to communicate with the system in a natural way.
(b) Explanation module : Gives the user, the ability to follow inferencing steps at any time during consultation.
(c) Interference engine : The interference engine is the main processing element of the expert system. The interference engine chooses rules from the agendas to fire. It there are no rules on the agenda, the internee engine must obtain information from the user in order to add more rules
(d) Knowledge base : Where the information is stored in the expert system in the form of fact & rules.

60. A computer based information system is needed :

(I) as it is difficult for administrative staff to process data
(II) due to rapid growth of information and communication technology
(III) due to growing size of organizations which need to process large volume of data
(IV) as timely and accurate decisions are to be taken

Which of the above statement (s) is/are true?

(a) I and II (b) III and IV
(c) II and III (d) II and IV

Ans : (b) A computer based information system is needed because of growing size of organizations which need to process large volume of data & so large data need timely & accurate decisions
So the correct choice is (b)

61. Given the recursively enumerable language (L_{RE}), the context sensitive language (L_{CS}), the recursive language (L_{REC}), the context free language (L_{CF}) and deterministic context free language (L_{DCF}). The relationship between these families is given by :

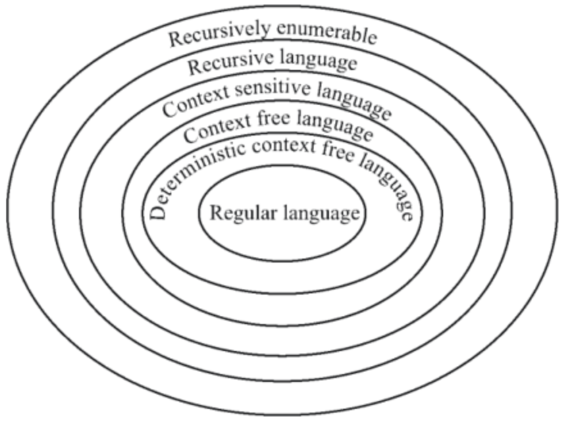
(a) $L_{CF} \subseteq L_{DCF} \subseteq L_{CS} \subseteq L_{RE} \subseteq L_{REC}$

(b) $L_{CF} \subseteq L_{DCF} \subseteq L_{CS} \subseteq L_{REC} \subseteq L_{RE}$

(c) $L_{DCF} \subseteq L_{CF} \subseteq L_{CS} \subseteq L_{RE} \subseteq L_{REC}$

(d) $L_{DCF} \subseteq L_{CF} \subseteq L_{CS} \subseteq L_{REC} \subseteq L_{RE}$

Ans : **(d)** Elements of the Chomsky Hierarchy



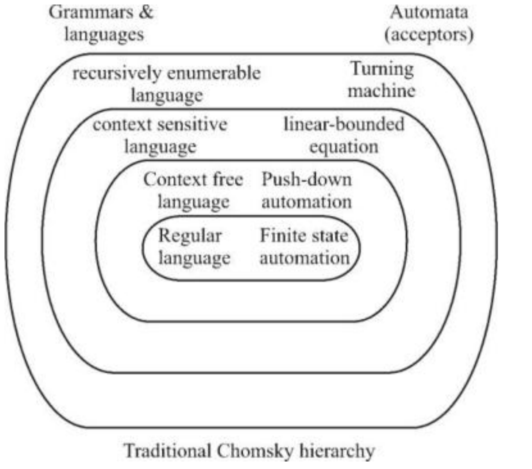
62. **Match the following :**

List –I		**List –II**	
A.	**Context free grammar**	**i.**	**Linear bounded automaton**
B.	**Regular grammar**	**ii.**	**Pushdown automaton**
C.	**Context sensitive grammar**	**iii.**	**Turing machine**
D.	**Unrestricted grammar**	**iv.**	**Deterministic finite automaton**

Codes :

	A	B	C	D
(a)	ii	iv	iii	i
(b)	ii	iv	i	iii
(c)	iv	i	ii	iii
(d)	i	iv	iii	ii

Ans : (b)



So the answer is (b)

63. According to pumping lemma for context free languages :

Let L be an infinite context free language, then there exists some positive integer m such that any $w \in L$ with $|w| \geq m$ can be decomposed as $w = u\,v\,x\,y\,z$

(a) with $|vxy| \leq m$ such that $uv^i\,xy^i\,z \in L$ for all $i = 0, 1, 2$

(b) with $|vxy| \leq m$, and $|vy| \geq 1$, such that $uv^i\,xy^i\,z \in L$ for all $i = 0, 1, 2,$ ....

(c) with $|vxy| \geq m$, and $|vy| \leq 1$, such that $uv^i\,xy^i\,z \in L$ for all $i = 0, 1, 2,$ ....

(d) with $|vxy| \geq m$, and $|vy| \geq 1$, such that $uv^i\,xy^i\,z \in L$ for all $i = 0, 1, 2,$ ....

Ans : (b) Context free pumping lemma to the following. Let L be an infinite context language. Then there exists some positive integer m such that any w that is a member of L with $|w| \geq m$ can be decomposed as $w = uvxyz$

with $|v \times y| \leq \& |vy| \geq 1$

such that $w_i = uv^ixy^i\,z$

is also in L for all i = 0, 1, 2, ...

64. Given two spatial masks

$$S_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{and} S_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The Laplacian of an image at all points (x, y) can be implemented by convolving the image with spatial mask. Which of the following can be used as the spatial mask?

(a) only S_1 (b) only S_2

(c) Both S_1 and S_2 (d) None of these

Ans : (b) Spatial mask for point (x, y) used to implement the digital laplacian of the 2-D is given by [f(x + 1, y), + (x –1, y), f(x, 4–1)–4 f(x, y)] so possible masks are.

$$\begin{matrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{matrix} \quad \text{or} \quad \begin{matrix} 1 & 1 & 1 \\ 1 & 8 & 1 \\ 1 & 1 & 1 \end{matrix}$$

here in S1 a 1 is mission on the right side of –4

So the correct answer is (b)

65. Given a simple image of size 10 × 10 whose histogram models the symbol probabilities and is given by :

p_1	p_2	p_3	p_4
a	b	c	d

The first order estimate of image entropy is maximum when

(a) a = 0, b = 0, c = 0, d = 1

(b) $a = \frac{1}{2}, b = \frac{1}{2}, c = 0, d = 0$

(c) $a = \frac{1}{3}, b = \frac{1}{3}, c = \frac{1}{3}, d = 0$

(d) $a = \frac{1}{4}, b = \frac{1}{4}, c = \frac{1}{4}, d = \frac{1}{4}$

Ans : **(d)** In option (d) all elements are in class 1/4. Hence entropy is least. In option c, a, 3 elements are in single class & one in other class. In option b, maximum diffusion, 2 in each class, hence max entropy means option d is the right answer.

66. A Butterworth lowpass filter of order n, with cutoff frequency at distance D_0 from the origin, has the transfer function H(u, v) given by :

(a) $\frac{1}{1+\left[\frac{D(u,v)}{D_0}\right]^{2n}}$ (b) $\frac{1}{1+\left[\frac{D(u,v)}{D_0}\right]^{n}}$

(c) $\frac{1}{1+\left[\frac{D_0}{D(u,v)}\right]^{2n}}$ (d) $\frac{1}{1+\left[\frac{D_0}{D(u,v)}\right]^{n}}$

Ans : **(a)** The transfer function of Butterworth low pass filter of order n with cutoff frequency at distance d_o the origin is defined as : H (u, v) = $1/1+(D(u, v)\ D_0)^{2n}$
So answer will be (a)

67. If an artificial variable is present in the 'basic variable' column of optimal simplex table, then the solution is :

(a) Optimum (b) Infeasible

(c) Unbounded (d) Degenerate

Ans : **(b)** Else if condition is false it may be unbounded
So answer is (b)

68. The occurrence of degeneracy while solving a transportation problem means that

(a) total supply equals total demand

(b) total supply does not equal total demand

(c) the solution so obtained is not feasible

(d) none of these

Ans : **(c)** Choice A is must be start for a solution
Choice B is converted to case A first then we try solution we can find unfeasible solution i.e. allocations are less than m+n–1 which is a case degenracy.

69. Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table :

		Jobs				
		I	II	III	IV	V
Men	P	2	9	2	7	1
	Q	6	8	7	6	1
	R	4	6	5	3	1
	S	4	2	7	3	1
	T	5	3	9	5	1

Find out the minimum time required to complete all the jobs.

(a) 5 (b) 11

(c) 13 (d) 15

Ans : (c) (By using Hungarian method)

Subtract smallest number from present in row/column

(1) All rows

(2) All columns

	I	II	III	IV	V
P	1	8	1	6	0
Q	5	7	6	5	0
R	3	5	4	2	0
S	3	1	6	2	0
T	4	2	8	4	0

⇒

0	7	0	4	0
4	6	5	3	0
2	4	3	0	0
2	0	5	0	0
3	1	7	3	0

0	7	0	5	0
3	5	4	3	0
1	3	2	0	0
2	0	4	1	0
2	0	6	3	0

⇒

0	8	0	5	1
2	5	3	2	0
1	4	2	0	0
1	0	4	0	0
1	0	5	2	0

0	0	0	6	2
1	5	2	2	0
0	4	1	0	1
0	0	3	0	0
0	0	4	2	0

now make assignment

P	$\cancel{0}$	9	$\boxed{0}$	6	2
Q	1	5	2	2	$\boxed{0}$
R	$\boxed{0}$	4	1	$\cancel{0}$	1
S	$\cancel{0}$	$\cancel{0}$	3	$\boxed{0}$	$\cancel{0}$
T	$\cancel{0}$	$\boxed{0}$	4	2	$\cancel{0}$

So, P–2, Q–1, R–4, S–3, T–3

= 2 + 1 + 4 + 3 + 3 = 13

So the answer is (c)

70. Consider the following statements about a perception :

I. Feature detector can be any function of the input parameters.

II. Learning procedure only adjusts the connection weights to the output layer.

Identify the correct statement out of the following :

(a) I is false and II is false
(b) I is true and II is false
(c) I is false and II is true
(d) I is true and II is true

Ans : (d) (i) Feature detection includes methods for computing abstractions of image information & it can be function of the input parameters
(ii) The learning procedure only adjust the connection to the output layer because output layer calculate it's output in the same way as the hidden layer.

71. A......... point of a fuzzy set A is point x ∈ at which $\mu_A(x) = 0.5$

(a) core (b) support
(c) crossover (d) α-cut

Ans : (c) The crossover of a fuzzy set is the element in U at which its membership function is o.5.
The support of a fuzzy set F is the crispset of all points in the universe of discourse U such that the membership function of f is non-zero
If membership is 1 then it is a core of fuzzy set alpha cut can be anything term to 1 (0<=λ⇐1)
μ A(x) = 0.5

72. Match the following learning modes w.r.t. characteristics of available information for learning :

A.	**Supervised**	**i.**	**Instructive information on desired responses explicitly specified by a teacher**
B.	**Recording**	**ii.**	**A priori design information for memory storing**
C.	**Reinforcement**	**iii.**	**Partial information about desired responses, or only "right" or "wrong" evaluative information**
D.	**Unsupervised**	**iv.**	**No information about desired responses**

Codes :

	A	B	C	D
(a)	i	ii	iii	iv
(b)	i	iii	ii	iv
(c)	ii	iv	iii	i
(d)	ii	iii	iv	i

Ans : (a) Supervise is under some guidance by some teacher
Reinforcement is evaluation in form of reward
Unsupervised is no information about desired responses
Recording is be some memory storage.

73. Which of the following versions of window O.S. contain built-in partition manager which allows us to shrink and expand pre-defined drives?

(a) Windows vista (b) Window 2000
(c) Window NT (d) Window 98

Ans : (a) Windows vista have logical disk manager that supports shrinking & expanding volumes vista also have a program that allow a user to upgrade their running vista to higher edition.

74. A Trojan horse is :

(a) A program that performs a legitimate function that is known to an operating system or its user and also has a hidden component that can be used for nefarious purposes like attacks on message security or impersonation
(b) A piece of code that can attach itself to other programs in the system and spread to other systems when programs are copied or transferred
(c) A program that spreads to other computer systems by exploiting security holes like weaknesses in facilities for creation of remote processes
(d) All of the above

Ans : (a) Trojan horse is any malicious program which is used to hack into a computer by misleading users of it true intent. Trojan can achieved any number of attack on the host from irritating the user to damaging the host (deleting files, stealing data, spread other virus)

75. Which of the following computing models is not an example of distributed computing environment?

(a) Cloud computing
(b) Parallel computing
(c) Cluster computing
(d) Peer-to-peer computing

Ans : (b) Parallel computing means that different activities happen at same time. It means speed out a single application over many processors to get it faster.